

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K} = (O, \mathcal{B})$ where $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

1. (1 point) Select all that apply. Which of the following statements are true?
 - A. Projections are invertible.
 - B. The homogeneous matrix of a transformation of $\mathbb{E}^3$ is a 4×4 matrix.
 - C. Reflections are invertible.
 - D. The tensor product of two vectors produces a symmetric matrix.

2. (1 point) The tensor product of the vectors $\overrightarrow{AB}(6, 0, 1)$ and $\overrightarrow{AC}(0, 2, 1)$ is $\overrightarrow{AB} \otimes \overrightarrow{AC} = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$.

Setup 1: Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

3. (1 point) Consider Setup 1 above. The orthogonal projection of the point C on the line AB is _____.
4. (1 point) Consider Setup 1 above. Relative to the frame in which the points are given, the matrix form of the orthogonal projection on the line AB is $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} - \\ - \end{bmatrix}$.
5. (1 point) Consider Setup 1 above. Relative to the frame $(A, \overrightarrow{AB}, \overrightarrow{AC})$, the reflection in the line AB parallel to $\overrightarrow{BC}$ of the midpoint of $[BC]$ is _____.
6. (1 point) Consider Setup 1 above. Relative to the frame $(A, \overrightarrow{AB}, \overrightarrow{AC})$, the matrix form of the reflection in the line AB parallel to $\overrightarrow{BC}$ is $\begin{bmatrix} x' \\ y' \end{bmatrix} \mapsto \begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} - \\ - \end{bmatrix}$.

Setup 2: Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

7. (1 point) Consider Setup 2 above. The orthogonal projection of the point D on the plane containing ABC is _____.
8. (1 point) Consider Setup 2 above. The reflection of the point D in the plane containing ABC parallel to $\overrightarrow{DA}$ is _____.
9. (1 point) Consider Setup 2 above. The orthogonal reflection of the point D in the line AB is _____.
10. (1 point) Consider Setup 2 above. Relative to the frame in which the points are given, the matrix form of the orthogonal reflection in the plane containing ABC is $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} - \\ - \\ - \end{bmatrix}$.
11. (1 point) Consider Setup 2 above. Relative to the frame $(B, \overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, the matrix form of the projection on the plane containing ABC parallel to $\overrightarrow{CD}$ is $\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \begin{bmatrix} - \\ - \\ - \end{bmatrix}$.

Answers:

1. B, C

$$2. \begin{bmatrix} 0 & 12 & 6 \\ 0 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

3. $(-3.5, 3.5)$

$$4. \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -7 \\ 7 \end{bmatrix}$$

5. $(1.5, -0.5)$

$$6. \begin{bmatrix} x' \\ y' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

7. $(-1.2, 1, 1.4)$

8. $(-2, -3, 0)$

9. $(-2, 1, 0)$

$$10. \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \frac{1}{5} \begin{bmatrix} -3 & 0 & -4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix}$$

$$11. \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$